Explainable AI for Lithography Hotspot Detection: A Multi-Method XAI Analysis

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Abstract*—Lithography hotspots are layout regions prone to printing defects such as opens and shorts caused by the interaction between lithography limitations and shrinking IC feature sizes. This work develops an explainable-AI (XAI) framework for lithography hotspot detection using CNNs trained independently on the five ICCAD-12 benchmarks. A lightweight reference CNN is compared with an XAI-ready variant that preserves spatial information through three named convolutional stages. The evaluation pipeline includes class-imbalance handling, leakage correction, baseline-versus-proposed comparison, and controlled ablations. Four attribution methods—Grad-CAM, Grad-CAM++, LayerCAM, and occlusion sensitivity—are applied to 57 representative predictions covering TP, TN, FP, and FN outcomes. Quantitative geometry-referenced tests and occlusion cross-checks show that attribution quality is method- and outcome-dependent: LayerCAM and occlusion provide the strongest consistent evidence, while Grad-CAM is useful mainly on true-positive hotspots and can become degenerate under gradient cancellation or model-output collapse. The core classification and multi-method XAI investigation are complete; remaining work concerns clean-split re-verification of flagged ablations, threshold calibration, and stronger physical/optical ground-truth validation.

***Index Terms*—Explainable artificial intelligence, lithography hotspot detection, convolutional neural network, Grad-CAM, Grad-CAM++, LayerCAM, occlusion sensitivity, ICCAD-12.**

I. INTRODUCTION

Photolithography is the dominant technique for patterning integrated circuits, but as feature sizes shrink below the lithography wavelength, certain layout patterns become susceptible to printing failures known as lithography hotspots. Typical failures include pinching that can produce open circuits and bridging that can produce short circuits.

Physics-based lithography simulation can identify such hotspots accurately, but exhaustive simulation is computationally expensive for full-chip verification. This motivates machine-learning based detectors that operate directly on layout clip images. The ICCAD-12 benchmark provides a standard basis for evaluating such detectors and contains five independent benchmark sets with severe and unequal class imbalance.

A CNN can provide a fast binary decision, but its black-box nature limits a designer's ability to understand why a layout was classified as a hotspot. The research question of this project is therefore: can explainable-AI techniques identify the layout regions or features that contribute most strongly to hotspot classification, and are those highlighted regions technically meaningful?

The project is organized into two connected stages. First, a lightweight CNN is trained independently on the five ICCAD-12 benchmarks for hotspot (HS) versus non-hotspot (NHS) classification. Second, the trained classifier is analyzed using multiple post-hoc attribution methods. The latest stage extends the original Grad-CAM plan to Grad-CAM++, LayerCAM, and occlusion sensitivity so that gradient-based explanations can be cross-checked against an independent forward-pass method.

The completed investigation contributes: (1) independent per-benchmark CNN training, (2) an XAI-ready architecture with preserved spatial feature maps, (3) explicit handling of severe class imbalance, (4) correction and quantification of a test/validation leakage bug, (5) a controlled imbalance ablation, (6) four-method XAI analysis across 57 representative predictions, and (7) geometry-referenced and occlusion-based validation of the explanations.

II. RELATED WORK

Verma et al. [1] compare VGG16-based transfer learning followed by SVM, Random Forest, or k-NN classification against a lightweight CNN trained from scratch on ICCAD-12. The lightweight CNN is reported to achieve higher balanced accuracy and faster inference, motivating its use as the reference baseline in this work. Borisov and Scheible [2] previously proposed a lightweight CNN design for lithography hotspot detection, while Liao et al. [3] and Yang et al. [4] studied transfer learning and shallow-to-deep learning approaches.

Class Activation Mapping (CAM) introduced class-discriminative localization using global-average pooling

[5]. Grad-CAM generalized this approach using gradients of a target-class score with respect to convolutional feature maps [6]. Grad-CAM++ [8] uses higher-order gradient information to improve localization, while LayerCAM [9] applies spatially varying gradient weighting at individual feature-map locations.

Occlusion sensitivity provides an independent attribution mechanism: input regions are systematically masked and the resulting change in the target-class score is measured. This makes occlusion useful for checking whether gradient-based maps reflect actual model sensitivity rather than gradient-specific artifacts.

LithoExp [7] is closely related because it addresses explainability and defect localization in lithographic hotspot detection. The present work differs by applying four post-hoc attribution methods to a classifier whose spatial representation is deliberately retained for XAI, and by quantitatively comparing the resulting maps against geometric layout proxies and occlusion sensitivity.

III. DATASET AND CLASS-IMBALANCE ANALYSIS

A. ICCAD-12 Dataset

All experiments use the five ICCAD-12 benchmarks, kept strictly separate for the primary results. Layout clips are loaded from the benchmark train_hs/train_nhs/test_hs/test_nhs partitions, resized to 224×224, and rescaled to [0,1]. Table I reports the image counts and the resulting class-imbalance ratios measured by the project pipeline.

Benchmark	Train HS	Train NHS	Test NHS:HS
ICCAD-1	99	340	20.7:1
ICCAD-2	174	5,285	82.9:1
ICCAD-3	909	4,643	25.6:1
ICCAD-4	95	4,452	180.2:1
ICCAD-5	26	2,716	471.4:1

TABLE I. PER-BENCHMARK CLASS DISTRIBUTION AND TEST-SET IMBALANCE RATIO.

Every benchmark is dominated by non-hotspot clips, with the test-split imbalance growing from roughly 20.7:1 on ICCAD-1 to 471.4:1 on ICCAD-5. A classifier that predicts NHS unconditionally could therefore achieve high raw accuracy while detecting no hotspots. Balanced accuracy, sensitivity, specificity, precision, and F1 are consequently emphasized.

The original project draft also recorded an ICCAD-1 test-count discrepancy. The current clean-split results use the dataset counts reported by the latest XAI pipeline; any remaining raw-dataset discrepancy should still be checked before final publication.

B. Preprocessing and Augmentation

For the latest XAI pipeline, layout clips are resized from their native 1200×1200 raster representation to 224×224 and rescaled to [0,1]. No rotation, flip, or other augmentation is used in the primary baseline or XAI-ready training runs, so both models operate on the same un-augmented per-benchmark train/test partitions.

C. Class-Imbalance Handling

Two imbalance strategies were investigated. The baseline uses no class weighting. The XAI-ready pipeline initially used inverse-frequency class weighting and selected checkpoints using balanced accuracy on a validation split. A dedicated ICCAD-5 ablation later showed that the loss-reweighted condition was responsible for a feature-collapse failure mode; balanced mini-batch sampling did not show the same collapse. Some ablation scripts still require clean-split re-verification, so those results are treated as provisional.

IV. METHODOLOGY

A. Baseline CNN

The baseline network reproduces the reference lightweight-CNN design [1], [2]. It uses stacked 3×3 convolutional layers with ELU activations and batch normalization, max-pooling, and a flatten-and-fully-connected classification head with dropout. With 224×224 input, the baseline reduces the spatial representation to approximately a 15×15 feature grid before flattening. This aggressive spatial reduction is important when interpreting the coarseness of its activation maps.

B. Proposed GAP-CNN

The proposed XAI-ready CNN keeps comparable convolutional depth but deliberately preserves spatial information for explanation. Three named stages are retained: conv_early_2 (112×112×16), conv_mid_2 (56×56×24), and conv_final_2 (28×28×32). The conv_final_2 feature map is the primary attribution target for Grad-CAM, Grad-CAM++, LayerCAM, and the corresponding cross-check analysis. The model then uses a late max-pool, flatten, Dropout(0.3), Dense(16, ReLU), and Dense(1, sigmoid) head.

Stage	Baseline CNN	Proposed XAI-ready CNN
Input	224×224×1 grayscale	224×224×1 grayscale
Spatial stages	Downsampled to ≈15×15	112×112×16; 56×56×24; 28×28×32
Convolution	Lightweight stacked 3×3 layers	Three named spatially-preserved stages
Classifier	Flatten + FC + Dropout(0.3)	Late pool + Flatten + Dropout(0.3) + Dense(16) + sigmoid
Attribution target	Coarse final representation	conv_final_2 (28×28×32)
Purpose	Reference baseline	Classification + multi-method XAI
Parameters	≈12.9k	≈25–35k

TABLE II. ARCHITECTURAL COMPARISON OF THE BASELINE AND PROPOSED XAI-READY CNN.

The architectural choice treats explainability as a design constraint: the classifier must retain spatially meaningful feature maps rather than discarding almost all location information before the classification head. This allows the same trained checkpoint to support multi-method explanation without changing the convolutional feature extractor.

C. Training Configuration

Both models are trained independently on the five benchmarks with batch size 32 and the Nadam optimizer. The baseline uses the fixed per-benchmark epoch budget reported in the reference implementation, while the XAI-ready model selects the best balanced-accuracy checkpoint using a clean validation split. The final test split is evaluated only after checkpoint selection. The latest pipeline therefore avoids the earlier test/validation leakage.

Parameter	Configuration
Optimizer	Nadam
Batch size	32
Input	224×224 grayscale
Baseline epochs	5 / 5 / 10 / 10 / 10
XAI checkpoint	Best validation balanced accuracy
Validation	15% stratified split from training data
Test use	Evaluated once after checkpoint selection

TABLE III. TRAINING CONFIGURATION FOR THE BASELINE AND XAI-READY CNN.

V. EVALUATION METRICS

With hotspot as the positive class, the confusion matrix contains true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). The standard metrics used throughout the project are:

$$\text{Precision} = TP / (TP + FP) \quad (1)$$

$$\text{Recall (Sensitivity)} = TP / (TP + FN) \quad (2)$$

$$\text{Specificity} = TN / (TN + FP) \quad (3)$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall}) \quad (4)$$

$$\text{Balanced Accuracy} = (\text{Sensitivity} + \text{Specificity}) / 2 \quad (5)$$

Balanced accuracy is treated as the primary summary metric because it gives equal importance to hotspot sensitivity and non-hotspot specificity regardless of class imbalance. Precision is important because false hotspot alarms can overload physical-design review, while sensitivity is important because missed hotspots represent potential manufacturing failures.

VI. EXPERIMENTAL RESULTS

A. Proposed GAP-CNN Results

Table IV combines the latest baseline and proposed XAI-ready results. The baseline achieves an average balanced accuracy of 0.9363, while the clean-split XAI-ready model achieves 0.9319. The small difference indicates that retaining spatial information for XAI does not materially change overall balanced accuracy, although the class-weighted XAI pipeline produces lower F1 on every benchmark because it trades precision for recall.

Benchmark	Baseline BA	XAI-ready BA	XAI Recall	XAI F1
ICCAD-1	0.8851	0.8743	0.9735	0.2937
ICCAD-2	0.9893	0.9841	0.9819	0.6281
ICCAD-3	0.9672	0.9508	0.9829	0.4837
ICCAD-4	0.8578	0.9534	0.9492	0.1983
ICCAD-5	0.9822	0.8969	0.8049	0.2292

TABLE IV. BASELINE AND PROPOSED XAI-READY CNN RESULTS ON THE CLEAN PER-BENCHMARK SPLITS.

The benchmark-level results show that both models maintain high specificity, but their precision/recall trade-offs differ. The proposed model retains strong hotspot recall, while its F1 values are lower on several benchmarks. This is a genuine limitation and motivates decision-threshold calibration using a validation split rather than the test set.

B. Baseline-versus-Proposed Comparison

The baseline-versus-proposed comparison is now available from the latest training runs. Table IV reports the complete per-benchmark comparison, replacing the earlier draft status table in which the baseline values were still pending.

Benchmark	Leaked BA	Clean BA	Δ (pp)
ICCAD-1	0.8935	0.8743	−1.92
ICCAD-2	0.9314	0.9841	+5.27
ICCAD-3	0.9641	0.9508	−1.32
ICCAD-4	0.9430	0.9534	+1.04
ICCAD-5	0.9501	0.8969	−5.32

TABLE V. EFFECT OF CORRECTING THE TEST/VALIDATION LEAKAGE BUG.

C. Cross-Benchmark Pooling Ablation

As a controlled ablation, a ResNet-18 backbone adapted to single-channel input was trained on all five ICCAD-12 benchmarks pooled into one training set and evaluated separately on each benchmark's test split. The pooled model produced unusually high detection rates and ROC-AUC values. Because pooling can increase repeated or near-duplicate layout motifs and the expanded evaluation data included transformed copies, these results are retained as a methodological ablation rather than the primary detector.

Benchmark	Test n	Detection Rate	False Alarm Rate	ROC-AUC
ICCAD-1	29,430	94.10%	1.82%	0.9897
ICCAD-2	250,776	98.53%	0.15%	0.9998
ICCAD-3	288,846	96.91%	1.47%	0.9930
ICCAD-4	192,402	93.50%	0.82%	0.9675
ICCAD-5	116,208	91.87%	0.20%	0.9994

TABLE VI. RESNET-18 TRAINED ON ALL FIVE BENCHMARKS POOLED TOGETHER AND EVALUATED PER BENCHMARK.

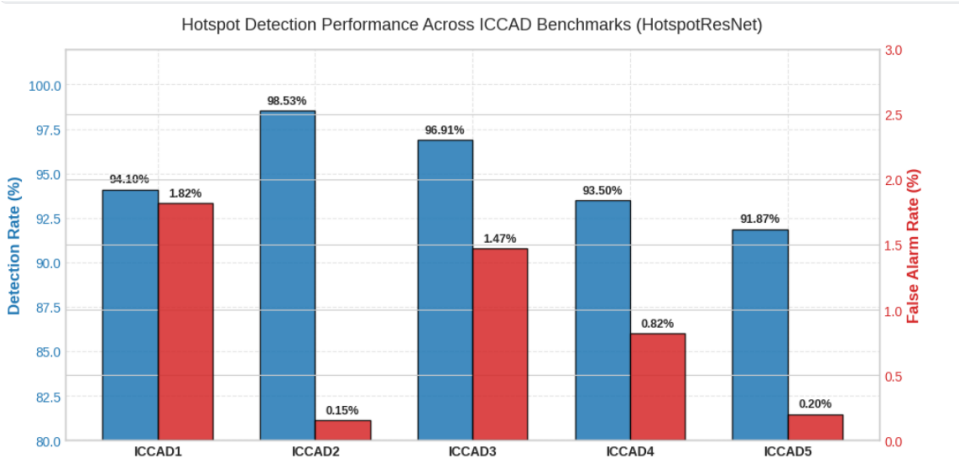


Fig. 1. Detection rate and false-alarm rate per benchmark for the pooled-training ResNet-18 ablation.

The pooled experiment reached detection rates above 91% and ROC-AUC values above 0.96 on all five benchmarks, but these numbers should not be interpreted as proof of superior generalization. The experiment demonstrates why dataset construction and train/test separation must be considered when interpreting very high performance.

The pooled test-set sample sizes were much larger than the raw partitions because each raw image was expanded into six orthogonal rotation/flip copies before the train/test boundary was preserved. This motivates strict separation of the five benchmarks for the main experiments.

The pooled-training design is therefore retained only as an ablation. The primary methodology uses independent benchmark training and evaluation, consistent with the severe class imbalance and the assignment requirement.

A threshold sweep performed directly on the pooled test set improved the reported ICCAD-1 detection rate slightly, but this is not used as a final result. Threshold selection must be performed on validation data to avoid optimistic test-set bias.

VII. ADDITIONAL EVALUATION AND PIPELINE VALIDATION

A. Pooled ResNet-18 ROC and Precision-Recall Curves

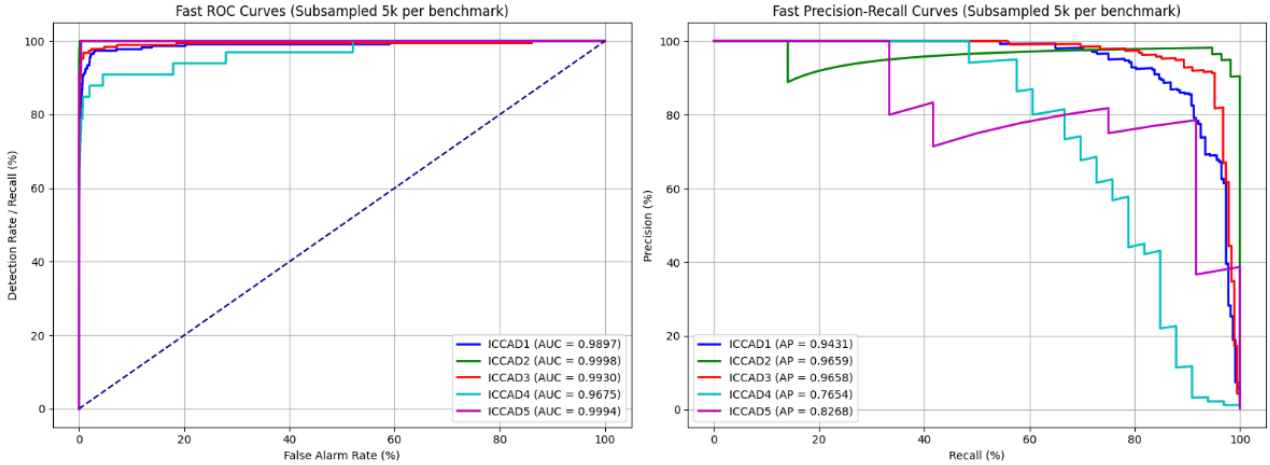


Fig. 2. ROC and precision-recall curves for the pooled-training ResNet-18 evaluated separately on each benchmark.

The ROC and precision-recall curves shown in the pooled ablation section correspond to the pooled ResNet-18 experiment rather than the primary XAI-ready CNN. They are retained to document the methodological finding that unusually strong pooled results require careful leakage and duplication checks.

B. Synthetic Pipeline Validation

Before trusting the evaluation pipeline on real benchmark data, the confusion-matrix computation, ROC/PR generation, and threshold sweep were exercised on a synthetic dataset constructed to resemble the ICCAD-1 class distribution. The purpose was to verify that the evaluation code produced internally consistent outputs before being applied to real data.

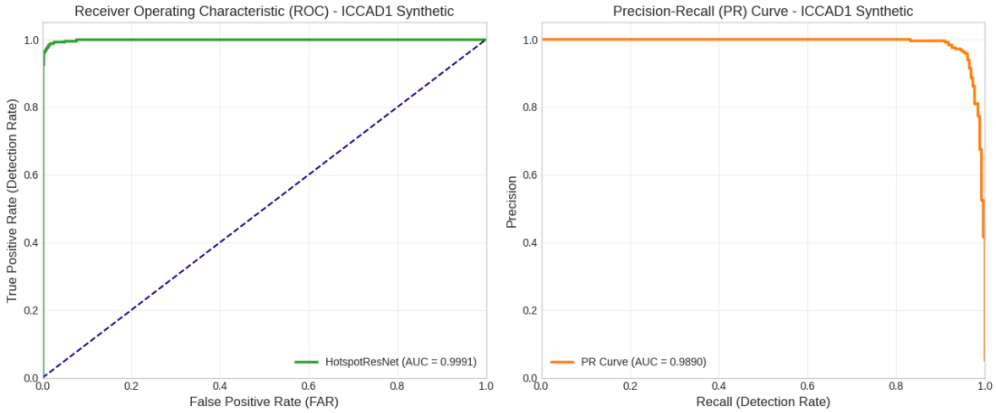


Fig. 3. ROC and precision-recall curves generated by the evaluation pipeline on the synthetic ICCAD-1-like validation set.

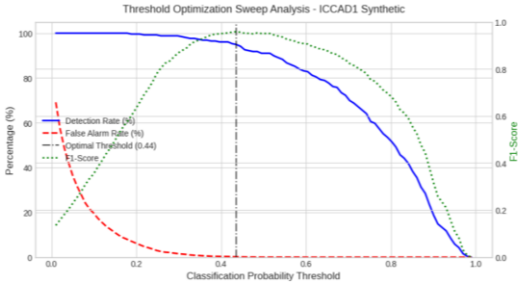


Fig. 4. Threshold-optimization sweep on the synthetic validation set; the optimum F1 threshold is shown at 0.44.

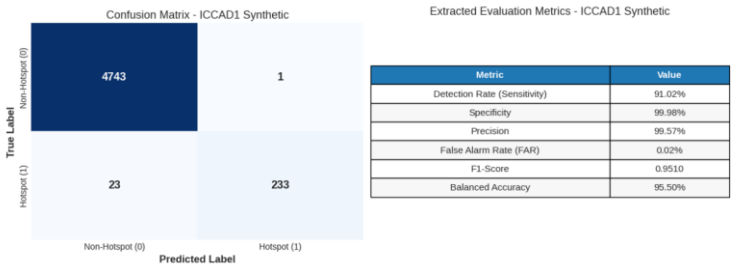


Fig. 5. Confusion matrix and extracted evaluation metrics from the synthetic validation run.

Metric	Value	Confusion-Matrix Basis
Detection Rate (Sensitivity)	91.02%	TP = 233
Specificity	99.98%	TN = 4,743
Precision	99.57%	FP = 1
False Alarm Rate	0.02%	FN = 23
F1-Score	0.9510	—
Balanced Accuracy	95.50%	—

TABLE VII. SYNTHETIC PIPELINE-VALIDATION METRICS. THESE ARE NOT REAL ICCAD-12 MODEL RESULTS.

The synthetic exercise confirms that the confusion-matrix, ROC/PR, and threshold-sweep code paths execute consistently. The resulting 95.50% balanced accuracy and 0.44 F1-optimal threshold belong only to the synthetic validation set and are not real ICCAD-12 detector performance.

The synthetic ROC analysis also reported an AUC of 0.9991. This value is a pipeline sanity-check result only and must not be interpreted as evidence about the real hotspot detector.

VIII. ABLATION STUDIES

A. Test/Validation Leakage Discovery and Correction

While preparing the explainability results, the XAI training pipeline was found to map the validation directory directly to each benchmark's test folder. This meant that checkpoint selection and final performance were computed on the same held-out data. The bug was corrected by carving a stratified 15% validation split from the training data with seed 42, leaving the test split for one final evaluation after training.

The clean-split comparison shows an average balanced-accuracy change of −0.45 percentage points, with a range from −5.32 to +5.27 percentage points. ICCAD-5 shows the largest reduction, consistent with its extremely small hotspot training population. The correction is treated as a methodological finding and the clean split is the basis for the current reported XAI-ready results.

B. Class-Imbalance Handling Ablation

A dedicated ICCAD-5 experiment compared three conditions while holding the architecture, optimizer, loss family, split, and seed fixed: no weighting, inverse-frequency class weighting, and balanced mini-batch sampling. Only the loss-reweighted condition exhibited catastrophic feature collapse, with 85.4% of hotspot-class penultimate activations becoming all-zero, whereas the non-reweighted conditions showed 0.0% collapse in the primary run.

A three-seed replication confirmed 0.0% collapse for balanced sampling, which is the condition adopted going forward. The original no-weight result did not replicate consistently across seeds, showing that it was seed-dependent. The class-imbalance ablation and separate seed-replication script still require clean-split re-verification and are therefore not treated as final claims.

Condition	Bal. Acc.	HS Recall	Zero-vector rate	HS-elevated units
baseline_noweight	0.9741	0.9512	0.000%	1 (AUC 0.982)
class_weighted	0.9649	0.9512	85.366%	0
balanced_sampling	0.9501	0.9024	0.000%	4 (AUC 0.987)

TABLE VIII. ICCAD-5 CLASS-IMBALANCE ABLATION. THE FEATURE-COLLAPSE EFFECT IS ATTRIBUTED TO LOSS REWEIGHTING IN THE PRIMARY RUN.

IX. EXPLAINABLE AI: MULTI-METHOD ANALYSIS

A. XAI-Ready Architecture and Attribution Setup

The final explainability study uses the proposed XAI-ready CNN and targets its conv_final_2 feature map (28×28×32). Four complementary methods are applied: Grad-CAM, Grad-CAM++, LayerCAM, and occlusion sensitivity. The first three are gradient-based CAM methods, while occlusion measures the change in the model output caused by masking local input regions and therefore provides an independent forward-pass reference.

The study covers 57 representative predictions from all five benchmarks: 15 true positives, 15 true negatives, 15 false positives, and 12 false negatives. This selection allows the explanation quality to be examined not only when the classifier is correct on hotspots, but also when it rejects non-hotspots or makes an error.

B. Representative Case Studies

The representative visualizations show a clear asymmetry. True-positive hotspot cases generally produce spatially coherent maps that overlap meaningful layout structures. In contrast, several true-negative cases show disagreement between CAM methods, border concentration, or degenerate maps. The following figures reproduce the completed visual evidence from the XAI analysis.

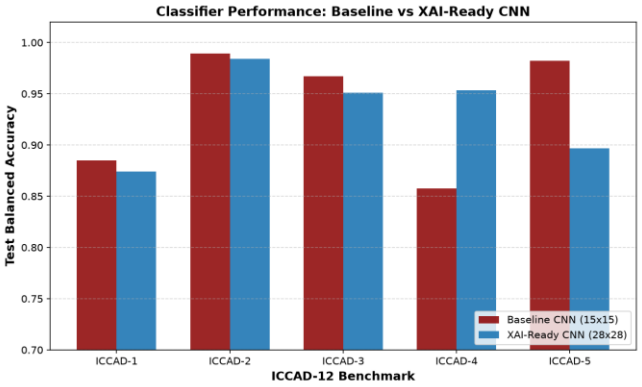


Fig. 6. Baseline versus proposed XAI-ready CNN balanced accuracy across the five ICCAD-12 benchmarks.

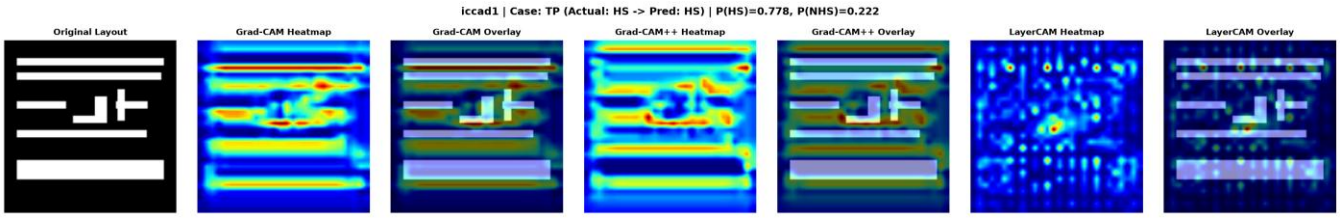


Fig. 7. ICCAD-1 HS123 (TP). Grad-CAM and Grad-CAM++ concentrate on a compact region, while LayerCAM provides a more distributed attribution; the highlighted area tracks a meaningful layout feature.

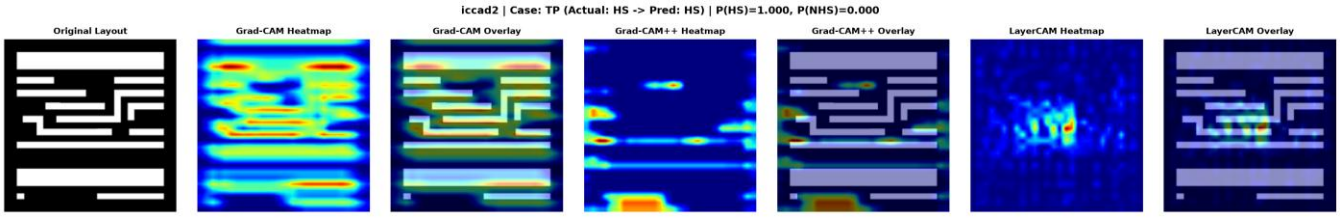


Fig. 8. ICCAD-2 HSCAD2146 (TP). The Grad-CAM, Grad-CAM++, LayerCAM, and occlusion evidence is strongly aligned; LayerCAM reaches a Pearson correlation of 0.758 with the independent occlusion map.

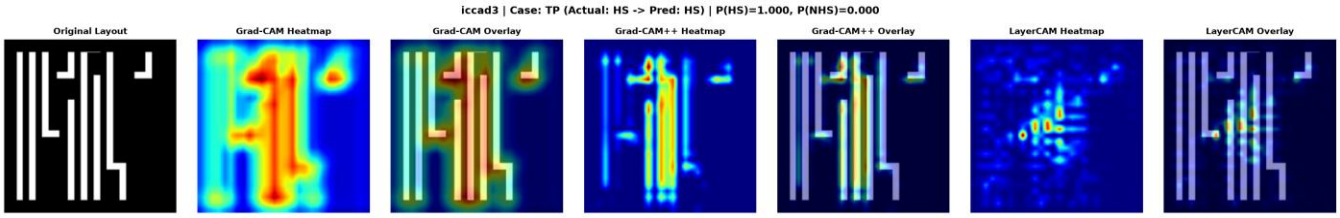


Fig. 9. ICCAD-3 HSCAD31012 (TP). All three CAM methods agree closely with the occlusion reference, with Pearson correlations of approximately 0.58, 0.59, and 0.75.

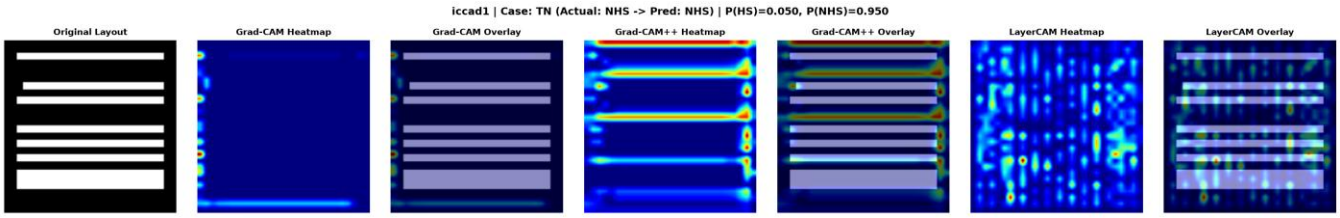


Fig. 10. ICCAD-1 NHS453 (TN). Grad-CAM is nearly degenerate and strongly concentrated near the image edge; the explanation is not clearly meaningful despite correct NHS classification.

C. Quantitative Geometry-Referenced Validation

Across the full 57-sample set, the top-20% attribution region of each method was compared with an area-matched random control using a paired Wilcoxon signed-rank test. The overlap measure used documented layout geometry including corners, junctions, and dense line regions. This turns the visual interpretation into a quantitative test rather than relying only on subjective heatmap inspection.

Method	Median / pooled effect	Significance	Interpretation
Grad-CAM	+0.033 overlap vs control	p=0.181 pooled; p=0.022 TP	Meaningful mainly on TP; below chance on TN
Grad-CAM++	+0.047 overlap vs control	p=0.016 pooled	Weak/variable; TP-only effect not significant
LayerCAM	+0.108 overlap vs control	p<0.001 pooled and TP/TN/FN	Consistent above-chance structure
Occlusion	+0.164 overlap vs control	p<0.001 pooled and TP/TN/FN	Strongest independent signal

TABLE IX. GEOMETRY-REFERENCED ATTRIBUTION RESULTS ACROSS 57 REPRESENTATIVE PREDICTIONS.

D. Grad-CAM and Grad-CAM++ Failure Modes

Two distinct failure mechanisms were verified. First, on selected ICCAD-5 samples, a direct gradient probe found an exactly zero gradient of the pre-sigmoid score with respect to conv_final_2, while the model produced the same hotspot probability and occlusion was also flat. This indicates genuine model-output collapse rather than a visualization-only bug. The affected checkpoint is pending clean-split re-migration.

Second, on ICCAD-3 sample NHSCAD31_7, the pre-sigmoid logit was saturated at approximately 21.1. About 75% of gradient elements were exactly zero and the

remaining elements nearly cancelled during Grad-CAM's global-average weighting, producing a flat map after ReLU. LayerCAM recovered spatial structure because it applies the gradient weighting pixel-wise instead of reducing each channel to one scalar.

Across all 57 samples, exact-flat-map rates were 31.6% for Grad-CAM, 42.1% for Grad-CAM++, and 8.8% for LayerCAM. These rates show why the project cannot rely on Grad-CAM alone.

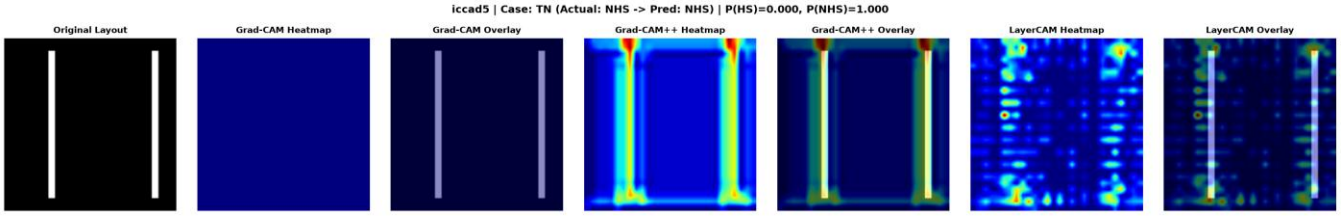


Fig. 11. ICCAD-5 NNHSCAD5999_6 (TN): Grad-CAM and Grad-CAM++ are flat in a model-output-collapse case; the checkpoint requires clean-split re-verification.

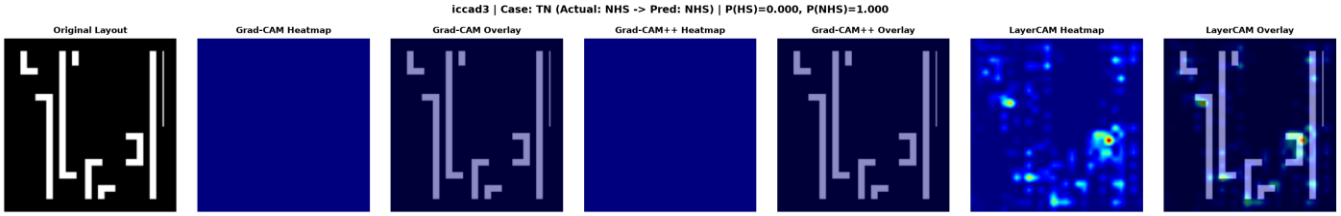


Fig. 12. ICCAD-3 NHSCAD31_7 (TN): Grad-CAM becomes flat under logit saturation, while LayerCAM retains structured spatial information.

E. Occlusion Cross-Check

Occlusion sensitivity provides an independent forward-pass reference. Across all 57 samples, the mean/median Pearson correlations with occlusion were 0.080/0.000 for Grad-CAM, 0.101/0.000 for Grad-CAM++, and 0.336/0.305 for LayerCAM. LayerCAM is therefore the only CAM method with a consistently positive and non-trivial median correlation. The disagreement of Grad-CAM and Grad-CAM++ with occlusion is itself evidence that gradient-based maps should not be interpreted in isolation.

F. Direct Answer to the Research Question

The evidence supports a qualified yes: XAI techniques can identify layout regions that plausibly contribute to hotspot classification, but the reliability depends strongly on the attribution method and prediction outcome. LayerCAM and occlusion sensitivity show statistically significant above-chance overlap with geometric layout features across true-positive, true-negative, and false-negative groups. Grad-CAM provides meaningful evidence particularly on true-positive hotspots, but it is chance-level when pooled, significantly below chance on true negatives, and degenerate on 31.6% of the analyzed samples.

X. DISCUSSION AND LIMITATIONS

The central finding is that explainability is not a single-method property of the classifier. The multi-method design is necessary because Grad-CAM and Grad-CAM++ can fail through gradient cancellation or model-output collapse, while LayerCAM and occlusion retain useful information on a broader set of outcomes.

The corrected test/validation split is another important methodological result. The original leakage bug changed balanced accuracy by −0.45 percentage points on average and by as much as −5.32 percentage points on ICCAD-5. This reinforces the need for validation-only checkpoint selection and one-time test evaluation.

The project does not contain validated optical-simulation or DRC ground truth for every highlighted region. Therefore, statements that an attribution is technically meaningful are based on geometric proxies such as corners, junctions, and dense-line structures rather than direct lithography simulation. This limitation should be addressed before making stronger physical claims.

The proposed model's F1 is lower than the baseline on every benchmark despite comparable balanced accuracy. The class-weighted loss therefore introduces a precision/recall trade-off that should be addressed by decision-threshold calibration using the clean validation split.

XI. CURRENT PROJECT STATUS AND FUTURE WORK

The project has progressed beyond the original 50% status reported in the earlier draft. The CNN classification stage is implemented and evaluated, the XAI-ready architecture is trained on the five benchmarks, the test/validation leakage has been identified and corrected, and the core

four-method XAI investigation has been completed on 57 representative predictions.

The completed XAI stage now includes Grad-CAM, Grad-CAM++, LayerCAM, and occlusion sensitivity, representative HS/NHS case studies, quantitative geometry-overlap testing, failure-mode analysis, and occlusion cross-validation. The results support a qualified answer to the research question rather than a blanket claim that every attribution map is reliable.

Remaining work is refinement rather than starting the XAI stage: (1) rerun the leakage-affected class-imbalance and seed-replication scripts on the clean split, (2) calibrate the decision threshold on validation data to improve precision/F1, (3) replace synthetic evaluation figures with complete clean-split real-data ROC/PR and confusion-matrix results where still missing, (4) extend geometric proxies toward validated optical/DRC ground truth, and (5) complete the GPU-dependent direct-gradient probe for the ICCAD-5 collapse mechanism.

XII. CONCLUSION

This This work combines CNN-based lithography hotspot detection with a **multi-method explainability framework**. A baseline CNN and an XAI-ready spatial-preserving CNN were evaluated on five ICCAD-12 benchmarks, while a leakage issue and feature-collapse failure were identified and corrected during development.

The completed XAI study applies **Grad-CAM, Grad-CAM++, LayerCAM, and occlusion sensitivity** to 57 representative predictions. **LayerCAM and occlusion** provided the strongest consistent evidence of meaningful layout localization, while **Grad-CAM** was effective for true-positive hotspots but showed failure cases.

The work will be continued by **improving and validating Grad-CAM**, particularly for gradient cancellation and output-collapse cases. Pending work includes **clean-split re-verification, threshold calibration, real-data ROC/PR analysis, gradient-probe validation, and stronger geometry-based validation against optical/DRC ground truth**.

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